THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL MCQ TEST SET 2

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. Which fraction is equal to 0.125? 1

(a) $\frac{1}{4}$ **(b)** $\frac{1}{12}$ **(c)** $\frac{1}{125}$ **(d)** $\frac{1}{8}$

2. A rule is: output $= 3\times$ input -4. The output for the input 9 is: 1

(a) 24 **(b)** 23 **(c)** 25 **(d)** 22

3. One angle of a linear pair is 2 times the other. The larger angle is: 1

(a) 140° **(b)** 120° **(c)** 110° **(d)** 130°

4. The number of reverse-and-add steps for 59 to become a palindrome is: 1

(a) 5 **(b)** 3 **(c)** 2 **(d)** 4

5. The two acute angles of a right triangle are in the ratio $1:2$. The smaller acute angle is: 1

(a) 20° **(b)** 30° **(c)** 40° **(d)** 50°

6. If $x \times \frac{3}{4} = \frac{9}{16}$, then $x \div \frac{3}{4}$ is: 1

(a) $1\frac{1}{2}$ **(b)** $\frac{1}{2}$ **(c)** 1 **(d)** 2

7. How many lakhs are there in 18 crore? 1

(a) 1810 **(b)** 1790 **(c)** 1800 **(d)** 2800

8. The value of $(7 + 3 \times (12 - 8)) \times 2 - 14$ is: 1

(a) 26 **(b)** 24 **(c)** 210 **(d)** 13

9. Which of these is the greatest? 1

(a) $\frac{1}{2}$ **(b)** $\frac{2}{5}$ **(c)** 0.405 **(d)** 0.45

10. If $x = 6$, the value of $2x^2 + 5$ is: 1

(a) 78 **(b)** 76 **(c)** 77 **(d)** 79

11. If $l \parallel m$, the bisectors of a pair of alternate interior angles are: 1

(a) parallel **(b)** coincident **(c)** perpendicular **(d)** intersecting

12. Which expression is a multiple of 9 for every digit pair (a, b) with $a > b$? 1

(a) $(10a + b) + (10b + a)$ **(b)** $a + b$ **(c)** $(10a + b) - (10b + a)$ **(d)** $10a + b$

13. The number of isosceles triangles with whole-number sides and perimeter 14 cm (equilateral included) is: 1

(a) 2 **(b)** 5 **(c)** 4 **(d)** 3

14. $\frac{5}{6} \div (\frac{5}{3} \div \frac{2}{3})$ equals: 1

(a) $\frac{1}{6}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{1}{3}$

15. The number of seconds in 30 days is: 1

(a) 25,92,000 **(b)** 2,59,200 **(c)** 25,88,400 **(d)** 25,93,000

16. The value of $250 \div (5 \times (7-2)) + 8 \times (6+3)$ is: 1

(a) 399 **(b)** 82 **(c)** 2139 **(d)** 84

17. $1.5 \times 0.2 \times 0.6 =$ 1

(a) 0.018 **(b)** 1.8 **(c)** 0.18 **(d)** 18

18. If $a = 4$ and $b = 5$, the value of $3a^2 - 7b$ is: 1

(a) 14 **(b)** 12 **(c)** 13 **(d)** 15

19. Two supplementary angles are in the ratio $2 : 3$. The larger angle is: 1

(a) 128° **(b)** 118° **(c)** 108° **(d)** 98°

20. The smallest 4-digit number divisible by 9 is: 1

(a) 1007 **(b)** 1009 **(c)** 1008 **(d)** 1017

21. The number of triangles with whole-number sides (in cm) and perimeter 12 cm is: 1

(a) 3 **(b)** 2 **(c)** 5 **(d)** 4

22. $\frac{3}{7}$ of a number is 12. Three times the number is: 1

(a) 42 **(b)** 84 **(c)** $84\frac{1}{2}$ **(d)** 168

23. The number of zeros in 1,25,000 × 80 is: 1

(a) 6 **(b)** 9 **(c)** 7 **(d)** 8

24. The value of 'twice the sum of 12 and 8, decreased by 15' is: 1

(a) 10 **(b)** 25 **(c)** 55 **(d)** 40

25. How many times does 0.25 go into 6? 1

(a) 23 **(b)** 24 **(c)** 26 **(d)** 25

26. Simplify $7(x+3) - 4(x-2)$: 1

(a) $3x + 29$ **(b)** $3x + 13$ **(c)** $3x - 13$ **(d)** $11x + 29$

27. The supplement of the complement of 35° is: 1

(a) 135° **(b)** 115° **(c)** 145° **(d)** 125°

28. The largest 4-digit number with all digits different that is divisible by 5 is: 1

(a) 9875 **(b)** 9874 **(c)** 9877 **(d)** 9876

29. The exterior angle at the base of an isosceles triangle is 125°. The apex angle is: 1

(a) 90° **(b)** 80° **(c)** 60° **(d)** 70°

30. A pipe fills $\frac{3}{4}$ of a tank in 15 minutes. The time to fill the whole tank (minutes) is: 1

(a) 20 **(b)** 40 **(c)** $20\frac{1}{2}$ **(d)** 10

31. 1 crore = 1,00,00,000; the number of crores in 4,50,00,000 is: 1

(a) 450 **(b)** 45 **(c)** 0.45 **(d)** 4.5

32. Compare 20 × 13 and 19 × 14. The first minus the second is: 1

(a) -6 **(b)** -7 **(c)** -4 **(d)** -5

33. 0.4 of a number is 3.2. The number is: 1

(a) 10 **(b)** 7 **(c)** 8 **(d)** 9

34. The perimeter of a rectangle with sides $7x + 6$ and $2x - 3$ is: 1

(a) $9x + 3$ **(b)** $28x$ **(c)** $10x + 18$ **(d)** $18x + 6$

35. The reflex angle that goes with an angle of $135°$ (so that together they make a complete turn) is: 1

(a) $225°$ **(b)** $235°$ **(c)** $215°$ **(d)** $245°$

36. Kaprekar's routine on 6321 reaches 6174 after how many steps? 1

(a) 6 **(b)** 7 **(c)** 8 **(d)** 9

37. The sum of the three exterior angles of a triangle (one at each vertex) is: 1

(a) $540°$ **(b)** $360°$ **(c)** $180°$ **(d)** $270°$

38. $\frac{2}{5}$ of a wall is already painted. Then $\frac{2}{3}$ of the remaining part is painted. The fraction of the whole wall still unpainted is: 1

(a) $\frac{1}{5}$ **(b)** $\frac{1}{10}$ **(c)** $\frac{2}{5}$ **(d)** $\frac{11}{30}$

39. A tank holds 3,50,000 litres. How many such tanks hold 1,26,00,000 litres? 1

(a) 35 **(b)** 36 **(c)** 37 **(d)** 46

40. The value of 45 × 99 can be found as 45 × 100 - 45 = : 1

(a) 4555 **(b)** 4455 **(c)** 4410 **(d)** 4500

41. $(0.35 + 0.65) \times 2.5 =$ 1

(a) 3.5 **(b)** 2.5 **(c)** 2.6 **(d)** 2.4

42. If $n = 14$, then $\frac{n(n+1)}{2}$ equals: 1

(a) 107 **(b)** 105 **(c)** 104 **(d)** 106

43. Two complementary angles differ by $30°$. The larger angle is: 1

(a) $70°$ **(b)** $60°$ **(c)** $80°$ **(d)** $50°$

44. The Collatz sequence from 15 needs how many steps to reach 1? 1

(a) 16 **(b)** 19 **(c)** 17 **(d)** 18

45. The angles of a triangle are $(3x)°$, $(2x + 10)°$ and $(4x - 10)°$. The largest angle is: 1

(a) $90°$ **(b)** $60°$ **(c)** $70°$ **(d)** $80°$

46. For which x is $x \div \frac{1}{3}$ less than x? 1

(a) $x < 1$ **(b)** no positive x

(c) every x **(d)** $x > 1$

47. The smallest number that must be added to 72,34,560 to get the next lakh is: 1

(a) 65,440 **(b)** 66,440 **(c)** 65,340 **(d)** 99,999

48. 15 × 47 + 15 × 53 is equal to 15 × ?. The missing number is: 1

(a) 99 **(b)** 115 **(c)** 100 **(d)** 101

49. Which of these is closest to 1? 1

(a) 0.93 **(b)** 1.04 **(c)** 1.1 **(d)** 0.8

50. Which expression is always odd for a whole number n? 1

(a) $2n + 4$ **(b)** $4n + 3$ **(c)** $n + 3$ **(d)** $4n + 2$

END OF QUESTION PAPER **Total: 50 Marks**

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